

Explanation Of Revisions

Title: Fight Fire with Fire: Adaptive Black-Box Defences for Misinformation Detection

ACL ARR 2026 May Submission 3459 -> August Resubmission

We would like to thank the reviewers and the area chair for the effort put into reviewing our contribution. While the paper in its current form is currently border-line recommended for acceptance into Findings, we are going to take the opportunity for further improvement.

A. Responses to Area Chair

Below you will find the key revisions suggested by the AC with our explanation on how we address them. At the end of this document we attach a version of our contribution including highlighted changes.

1. Estimating true label for reinforcement learning reward

As reviewers M6fk and eKBm point out, our initial MACABEU-online method was limited by the fact that the reward function depended on the true gold label, which might not be known in practice. Therefore, we compare the offline approach to:

- MACABEU-oracle, using the true gold label as previously, if it is available for adaptation, for example coming from human feedback.
- MACABEU-estimated, where the true label is estimated using 7-fold voting (one of the defense mechanisms). This allows us to proceed even if the gold label is unknown.

In section 3.3 we describe the two new MACABEU variants. We have added a new Table 9 in Appendix E, comparing their performance on BiLSTM and BERT. The estimated variant preserves most of the online adaptation gain: mean BODEGA is only ~20% higher than oracle on the BiLSTM+BERT sub-grid, and still below every static defence in Table 1. On character-level attackers (BERTattack, XARELLO) it comes within 5% of oracle; on word-level ones (PWWS, Genetic) the gap is wider (+60% and +27%), a known limitation of consensus-based pseudo-labels that we discuss in Limitations (x). We believe this is a valuable improvement, making the proposed approach directly applicable.

2. Type of RL paradigm

As reviewer eKBm pointed out, we claimed to have used Q-learning, but the actual method employed (contextual bandit) does not fall under this paradigm, as it disregards future rewards.

This was indeed a mistake, and it has been corrected. The relevant section (3.3) was amended accordingly.

3. Justification of query capping

Our experiments limit the number of queries to 500, making it difficult to attack examples with longer text. This aspect was commented on by reviewer LQQa.

While we acknowledge it makes attackers' role more difficult, we need to limit the query budget to make the procedure realistic. No public platform would allow its users to make thousands of attempts to publish a post. This limitation has also been introduced in previous work, with even lower quotas than ours. We now provide a justification and reference to relevant work in section 3.1.

4. MACABEU feature importance

MACABEU uses a variety of features to decide which defence is appropriate for a particular example. As reviewer M6fk notes, we do not perform experiments to find out which of those are indeed relevant for the task.

To alleviate this limitation, we analyse the contribution of each of them by checking how they influence the choices made by MACABEU. The whole experiment and its results are described in appendix F, with a shortened conclusion in section 5.3. Four features dominate the policy's decisions across all trained variants: text length, uppercase ratio, word count, and OOV ratio; the remaining six features have flip rates below 0.11 and barely influence action selection.

5. Framing problem

Indeed, the fragments of the introduction that made it look like there is a "lack of black box defense methods" and we address it by designing a new technique, might appear misleading. We have clarified it by rewording several fragments, especially the contributions at the end of the introduction section in order to make it clear that we provide:

- A systematic analysis of how the already existing black-box defences fare in the misinformation detection use cases,
- A meta-defender that combines these defences to find the best approach for each adversarial example.

In the related work section we also review a plethora of previous publications on black-box defences.

B. Other Changes

Apart from the changes above, we have improved the manuscript in the following ways:

6. Availability of source code

As reviewer eKBm pointed out, the anonymised source code link made it impossible for the reviewers to access it. In order to make the code available while keeping the anonymity requirements, we attach the code to the submission.

7. Clean accuracy scores

Our measurements of clean accuracy (when no attack is performed) in Table 2 and Figure 3 included static defences and MACABEU-off. Reviewer M6fk pointed out that the results should also cover MACABEU online.

This was improved by adding the two online variants (MACABEU-oracle and MACABEU-estimated) to the clean accuracy results. Both are competitive: MACABEU-oracle costs only -0.6% clean accuracy relative to baseline while cutting mean BODEGA to 0.097, and MACABEU-estimated costs -4.0% while still dominating every static ensemble defence on the accuracy–robustness plane (Figure 3).

8. Other changes

We have also corrected further small mistakes and improved editing to fit in the page limit despite expanding the content to address the reviewers’ concerns.

9. Lack of analysis of cross-architecture differences, larger victim models or attackers adapting to online MACABEU

We are sorry, but the limitations of both the length of this paper (which already includes numerous appendices providing additional information), as well as the extensive computational cost of training attackers and defenders, mean that these additional experiments will have to remain out of scope. Our study provides the first comparison of black-box defences in literature: systematically across tasks, victim architectures, and attackers, as well as a learned per-example defence selector. We hope to explore the further extensions to this seminal study in future work.

D. Change-Highlighted Version of the Manuscript

(see below)

Fight Fire with Fire: Adaptive Black-Box Defences for Misinformation Detection

Anonymous ACL submission

Abstract

Text classifiers used to filter misinformation are vulnerable to adversarial examples, but most existing defences require white-box access to the victim model. We present the first systematic comparison of black-box preprocessing defences for credibility assessment, evaluating twelve techniques across four tasks, three victim architectures, and five attackers — including the adaptive reinforcement-learning attacker. We then introduce MACABEU, a reinforcement-learning meta-defender that selects which static defence to deploy for a given example, in both an offline (frozen-policy) and an online (continually-updated) regimes. Our experiments show that simple static defences, when chosen well, already neutralise non-adaptive attackers; that adaptive attackers re-establish a substantial threat against any single static defence; and that an adaptive defender recovers most of the lost robustness without modifying the victim. The best MACABEU variant reduces the efficiency of the attacks by 10–40% relative to static defences, with improvements on every one of 12 task–victim combinations, even against an adaptive attacker.

1 Introduction

Text classification models based on neural networks have been increasingly popular due to high decision accuracy, but they suffer from vulnerability to *adversarial examples* (AEs). This term describes pieces of data (i.e. text fragments) that have been purposefully modified by an attacker to elicit incorrect decisions from a victim classifier (Zhang et al., 2020). Credibility assessment of user-generated text, e.g. within content filtering for a social network, is a good example where robustness to such attacks plays a key role (Carter et al., 2021).

Previous research (Przybyła et al., 2024b) has shown that, indeed, often a small change made to

the non-credible input (e.g. a fake-news article) can confuse a misinformation detector to rate it as credible. AEs are especially effective when the attacker is *adaptive*, i.e. has had the opportunity to learn the victim’s vulnerabilities through previous interaction (Przybyła et al., 2024a).

Less attention has been given to how to design *defences* against such attacks. Classical defences modify the victim (e.g. adversarial training, loss changes; Goyal et al., 2023), but most deployed classifiers are LLMs served as APIs, whose users cannot touch the weights or the training loop.

Therefore, in this work we investigate available *black-box defences*: techniques that make the victim classifier more robust without any changes to the model itself. Specifically, we answer the following research questions in the context of misinformation detection:

- RQ1:** Which existing black-box defences provide the best robustness across attacks, text domains, and victim classifiers?
- RQ2:** How do these static defences fare against an adaptive attacker — both one pre-trained on the original victim and re-trained against a known defence?
- RQ3:** Can a defender itself be made adaptive, choosing the most effective defence per example or even continuing to learn during the attack?

To address RQ3 we introduce **MACABEU** (**MA**king **C**redibility **A**ssessment **r**oBust when **E**valuated **U**nder attacks), a reinforcement-learning meta-defender that selects, on a per-example basis, which existing defence from a fixed pool to deploy. MACABEU operates in an *offline* mode, in which the policy is frozen at deployment, and *two online* modes in which the policy continues to update from real attacker queries observed at evaluation time.

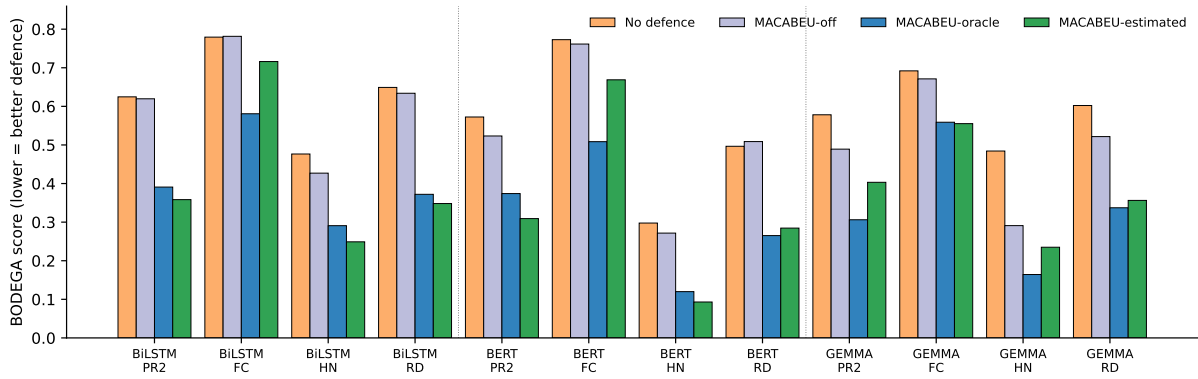


Figure 1: **Headline result.** Attack success (BODEGA score) of the pre-trained adaptive attacker against an undefended victim and with offline or online (estimated or oracle) MACABEU defender, across four tasks and three victims.

The latter are, to our knowledge, the first defenders from textual AEs that adapt *during* an attack.

To encourage further research in this direction, we share the source code used for these experiments¹.

2 Related Work

Adversarial examples for text. Since the seminal observation that small, semantics-preserving perturbations can fool neural classifiers (Goodfellow et al., 2015; Szegedy et al., 2014), a large body of work has explored AE generation for text. Attacks operate at different granularities: character-level (DeepWordBug (Gao et al., 2018), VIPER (Eger et al., 2019)), word-level (TextFooler (Jin et al., 2020), BERT-Attack (Li et al., 2020), PWWS (Ren et al., 2019)), and sentence-level paraphrasing (Iyyer et al., 2018). Search strategies range from greedy saliency-based replacement to gradient-free black-box optimisation such as genetic algorithms (Alzantot et al., 2018) and particle swarm (Zang et al., 2020). Recent work has explored LLM-driven attacks that generate fluent rewrites in a single pass (Moraffah et al., 2024; Zou et al., 2023). Toolkits such as OpenAttack (Zeng et al., 2021) and TextAttack (Morris et al., 2020) make these methods readily reproducible.

Robustness of misinformation detectors. Text credibility assessment (Srba et al., 2026) is an application in particular danger of AEs, e.g. an attacker who tricks a fake-news classifier can disseminate manipulated content at scale. Przybyła et al. (2024b) introduced BODEGA, a bench-

mark covering four credibility tasks (propaganda, fact-checking, hyperpartisan news, rumour detection) and a composite metric balancing attack success with semantic and character-level similarity. Follow-up work has shown that adaptive attackers, which learn the victim’s behaviour during interaction, can be substantially more effective than fixed heuristics; XARELLO (Przybyła et al., 2024a) formalises this as reinforcement learning over a bandit of perturbation operators with a Transformer-based Q -estimator. Subsequent work has extended this picture to LLM-generated adversarial inputs against fake-news and fact-checking classifiers (Bethany et al., 2025; Aïdahoul and Zaki, 2025; Przybyła et al., 2025), and recent surveys catalogue both the attack surface for automated fact-checking (Liu et al., 2025) and the broader credibility-signal landscape under LLMs (Srba et al., 2026), confirming that credibility classifiers remain brittle outside the original BODEGA setup.

Threat models. The two most important aspects of the threat model are the level of access to the victim model enjoyed by the attackers and defenders. In terms of the **attacker’s** access, the *white-box* model assumes full access to the victims’ weights, allowing gradient attacks such as FGSM (Goodfellow et al., 2015) and PGD (Madry et al., 2018). However, it is not realistic to assume such access on the attackers’ part. Previous work in misinformation domain focused on the scenario labelled *black-box* (only final decisions are visible) or *grey-box* (a continuous probability score is visible). Concrete examples include label-only black-box search such as PWWS (Ren et al., 2019) and the genetic attack of Alzantot et al. (2018), and score-based grey-box

¹Attached to the submission for review; when published available at [https://github.com/\[ANONYMOUS\]/macabeu](https://github.com/[ANONYMOUS]/macabeu)

attacks such as DeepWordBug (Gao et al., 2018), TextFooler (Jin et al., 2020) and BERT-Attack (Li et al., 2020). These better match the realistic deployment of credibility filters as third-party APIs. We base our work on the grey-box approach and include various modification of the output score (including black-boxing) as defence techniques. In terms of **defender’s** access, we also assume the *black-box* model, treating the victim as a third-party API.

Defences against textual AEs. Defensive techniques fall into three broad families (Goyal et al., 2023; Qiu et al., 2022). (i) **White-box training-time defences** modify the victim via adversarial training (Madry et al., 2018; Miyato et al., 2017), certified robustness (Jia et al., 2019; Ye et al., 2020), or specialised loss terms (Zhu et al., 2020) — all impractical for users of large hosted models. (ii) **Detection-based defences** flag suspect inputs rather than correcting them (Mozes et al., 2021; Zhou et al., 2019), at the cost of latency and false positives. (iii) **Black-box wrap-around defences** transform either the input or the output of the victim before it reaches downstream consumers. On the *input side*, examples include spell correction (Pruthi et al., 2019), randomised smoothing (Ye et al., 2020), character-level denoising (Jones et al., 2020), and embedding-space robustification against claim edits (Magomere et al., 2025); recent work has extended this strand to LLM-based input purification (Moraffah et al., 2024; Robey et al., 2025) and to adversarial-aware multi-agent filtering pipelines (AIDahoul and Zaki, 2025; Tong et al., 2025). On the *output side*, post-processing the classifier’s probability vector — argmax discretisation (returning only the predicted label), additive output noise, stochastic decision thresholds, or random label flips — demotes the attacker’s interface from grey-box to black-box at zero training cost. Such output-randomisation tactics have been analysed both as a generic obfuscation defence and as a target for adaptive attackers (Tramèr et al., 2020; Athalye et al., 2018). This family is attractive because it requires no model access and can be wrapped around any classifier; we revisit and benchmark four output-side variants in Section 3.2. The CLEF-2024 CheckThat! lab (Piskorski et al., 2024) likewise added an adversarial-robustness track for credibility tasks, mirroring our motivation.

To our knowledge, no prior work in text classification has compared black-box defences systemati-

cally across tasks, victim architectures, and *adaptive* attackers, nor combined them with a learned per-example defence selector. The closest prior benchmark (Li et al., 2021) evaluates 11 word-substitution defences on classification and sentiment tasks, but covers only non-adaptive attackers and fixed defences. We close both gaps here, in the credibility-assessment setting.

3 Methods

3.1 Attack evaluation framework

We adopt the BODEGA evaluation protocol (Przybyła et al., 2024b). The benchmark covers four English-language credibility tasks — propaganda detection (**PR2**), fact-checking (**FC**), hyperpartisan news detection (**HN**), and rumour detection (**RD**) — each detailed in Section 4. Each task provides labelled training data and an evaluation set of inputs that the victim classifies correctly in the absence of attack. The attacker queries the victim $f(\cdot)$ and produces a perturbed input x' from each original x . A trial is scored along three axes:

- **Attack success** — whether the victim’s prediction flipped, i.e. $f(x) \neq f(x')$ (binary, averaged into the attack success rate, ASR);
- **Semantic similarity** between x and x' , measured with BLEURT (Sellam et al., 2020);
- **Character similarity** between x and x' , computed as the normalised Levenshtein distance (Levenshtein, 1966).

The composite *BODEGA score* is the product of these three components, rewarding attacks that are simultaneously effective and unobtrusive. We measure attacker effectiveness by the average BODEGA score on each task–victim–defence–attacker combination; a failed attack scores zero. For each defence we also report clean accuracy on unattacked inputs. We cap each attack at 500 victim queries².

3.2 Static defences

A static defence is a deterministic or stochastic transformation inserted in the victim’s input ($t_{\text{in}}(\cdot)$) or output ($t_{\text{out}}(\cdot)$), replacing $f(\cdot)$ with

²This reduces the performance of some attackers, especially on longer text. However, it would be unrealistic to assume unlimited queries to the attacked service and previous research explored even stricter limits (Przybyła et al., 2025)

$t_{\text{out}}(f(t_{\text{in}}(\cdot)))$. We evaluate four mechanism families; Table 5 (Appendix B) shows each defence applied to the same example.

Input cleaning: **spellcheck** corrects OOV tokens against a SymSpell frequency dictionary; **unicode** applies NFKC normalisation, ASCII-maps confusable codepoints, and strips zero-width characters — a direct counter to homoglyph attacks.

Input perturbation: **char_noise** _{σ} substitutes each character with a Unicode homoglyph with probability σ , on the assumption that an attacker’s crafted edits are more fragile to extra noise than natural language is.

Output masking: **discretize** replaces the victim’s probability vector with a one-hot, hiding the soft scores attackers exploit to estimate gradients (turning the attack into a black-box scenario).

Smoothing and combinations: **majority_vote@ k** draws k stochastic copies of the input and aggregates the victim’s verdicts on them. Each copy is produced by independently perturbing every non-whitespace character with probability $p=0.1$, applying one of three edits chosen uniformly at random: deletion, swap with the next character, or insertion of a random lowercase letter. Rather than collapsing the k verdicts to a hard label, we return the vote-fractions as a probability vector, so the API still exposes a continuous score but one that is quantised to $1/k$ and re-sampled afresh on every query — a noisy oracle in the spirit of randomised smoothing (Ye et al., 2020). **spellcheck_mv@ k** composes spellcheck with this majority vote. We test $k \in \{3, 7\}$.

For each defence we report both clean accuracy (Table 2) and robustness against the four standard attackers (Table 1), making the trade-off explicit.

3.3 Adaptive defences (MACABEU)

Static defences are brittle once the attacker can adapt to a known transformation: Athalye et al. (2018) and Tramèr et al. (2020) reduce, respectively, 7/9 and 13/13 published defences to baseline robustness with defence-specific attacks, and our RQ2 experiment (Section 5.2) reproduces this for text. Yet the defender does not, in practice, know which attack a given example uses. We therefore let MACABEU pick which static defence to apply per example, framing this as a **contextual bandit** (Li et al., 2010) acting on input features, similar to XARELLO’s attacker-side construction (Przybyła et al., 2024a). Figure 2 sketches the resulting loop.

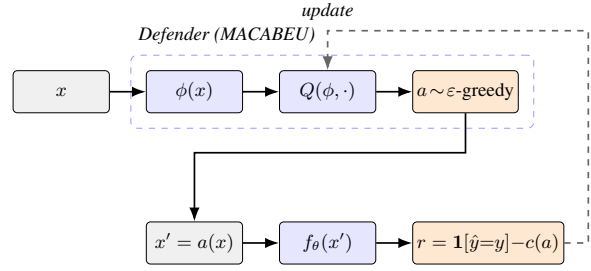


Figure 2: MACABEU as a defender-side RL loop.

Formulation. We frame defence selection as a contextual bandit (Li et al., 2010; Langford and Zhang, 2007): each evaluation example is a single-step episode in which the agent observes a feature vector $\phi(x)$ of the input x , picks one defence $a \in \mathcal{A}$ to apply to it, and receives a scalar reward determined by whether the victim’s prediction on the defended text matches the gold label.

The state $\phi(x)$ is a 10-dimensional vector of surface statistics: length, word count, average word length, OOV ratio, non-ASCII ratio, uppercase ratio, digit ratio, punctuation density, repeated-character ratio, and character entropy. Each is a known signature of a different attack family, so $\phi(x)$ carries enough signal for the policy to infer the attack and the defence likely to undo it.

The action space $\mathcal{A}$ contains eight defences from Section 3.2, in the four mechanism families: none (pass-through), spellcheck, unicode, majority_vote@3, majority_vote@7, spellcheck_mv@3, discretize, and char_noise@0.10; Table 5 (appendix B) shows each applied to the same example.

The reward is $r(a) = +1$ if the victim’s prediction on the defended input matches the gold label and -1 otherwise, minus a small inference-cost penalty $c(a)$ that discourages over-using the most expensive defences ($c = 0.10$ for $k=7$ smoothing, 0.05 for $k=3$, 0.02 for char_noise, 0 otherwise). The action-value network is a 3-layer MLP (64/32 hidden, $\sim 3k$ parameters) trained with ϵ -greedy exploration and Adam (Kingma and Ba, 2015).

We consider three operating regimes:

Offline (MACABEU-off). The policy is trained on synthetic attacker–defender interactions over the training portion in BODEGA: for each training example we simulate a randomly chosen attacker, observe the outcome under each candidate defence, and update the action-value network. At deployment (attack portion) the policy is frozen.

Defence	BERTAttack	PWWS	DeepWordBug	Genetic	XARELLO	Mean
none	0.445	0.357	0.259	0.376	0.585	0.405
spellcheck	0.448	0.366	0.087	0.344	—	0.311
unicode	0.467	0.375	0.182	0.374	—	0.349
discretize	0.295	0.262	0.178	0.228	—	0.241
MV@3	0.138	0.143	0.111	0.146	—	0.135
MV@7	0.128	0.148	0.107	0.165	—	0.137
sc_mv@3	0.127	0.130	0.075	0.159	0.374	0.173
char_noise@0.1	0.142	0.148	0.100	0.163	—	0.138
MACABEU-off	0.178	0.177	0.102	0.189	0.542	0.238
MACABEU-oracle	0.102	0.116	0.060	0.111	0.356	0.149

Table 1: Mean BODEGA per (defence, attacker), averaged over 4 tasks $\times$ 3 victims. Lower = stronger; best per column in **bold**. ‘—’ = pair not evaluated. MACABEU-oracle is a theoretical upper bound (it uses the gold label at inference time); its deployment-realistic counterpart, MACABEU-estimated, is compared against it in Appendix E.

Online. The offline policy is used as a warm start, and the agent continues to update its network from real attacker queries observed at evaluation time, using a replay buffer of size 16 with a 50-example warm-up. The defender thus adapts to the specific attacker it is facing, mirroring the asymmetry assumed by adaptive attackers. **This process requires access to true labels in order to signal the effectiveness of the defence. We consider two variants: assuming true labels are available during adaptation, e.g. from humans in the loop (MACABEU-oracle), or estimated by checking the victim’s response to majority_vote@7 (MACABEU-estimated).** A direct comparison of the two variants on the subset of task–victim cells where the estimated variant finished within the revision budget is presented in Appendix E; unless stated otherwise, MACABEU-on in the main text refers to the oracle variant to ensure comparability across the full 3-victim grid.

4 Experiments

Tasks and data. We use the four tasks of text credibility assessment in English bundled with BODEGA (Przybyła et al., 2024b): propaganda detection (PR2; based on SemEval-2020 Task 11 (Martino et al., 2020)), fact checking (FC; FEVER (Thorne et al., 2018)), hyperpartisan news detection (HN; PAN at SemEval-2019 (Kiesel et al., 2021)), and rumour detection (RD; augmented RNR (Zubiaga et al., 2015)). Their training sizes differ greatly ($\sim 8.7k$ for RD up to $\sim 173k$ for FC; full per-split counts in Table 4, appendix A).

Victim classifiers. To probe whether defences transfer across architectures, we train three victims per task: a **BiLSTM** over 300-dimensional GloVe embeddings, a fine-tuned **BERT-base**, and a QLoRA-fine-tuned **Gemma-2B**. They differ in

parameter count and cover the dominant paradigms in current text classification systems.

Attackers. We re-use the four standard black-box attackers and the adaptive attacker XARELLO introduced in Section 2; BERT-Attack, PWWS, DeepWordBug, Genetic, XARELLO, each capped at 500 victim queries per example.

Protocols. We conduct three experimental blocks. **RQ1:** the twelve static defences against the four standard attackers on every task–victim combination (Gemma restricted to the eight strongest defences identified on BiLSTM and BERT to limit computational resources), with clean-accuracy cost. **RQ2:** XARELLO against the strongest static defences on every combination, isolating attacker adaptivity. **RQ3:** MACABEU variants against all five attackers on every combination, with GEMMA excluded for MACABEU-estimated (see Appendix E) — isolating defender adaptivity (vs. RQ2) and the gain of an adaptive defender over the best static defence (vs. RQ1).

Implementation. Hardware, hyperparameters, and library versions are reported in Appendix C.

5 Results

5.1 RQ1: Static defences

Aggregated robustness. Table 1 reports the mean BODEGA score for each (defence, attacker) pair, aggregated over all 12 task–victim sub-combinations (4 tasks $\times$ 3 victims). Focusing on the four standard attacker columns (BERT-Attack, PWWS, DeepWordBug, Genetic), three clear patterns emerge.

(i) **Spellcheck foils character-level attacks but misses word-level ones.** Against DeepWord-

Defence	PR2	FC	HN	RD	Mean
none	0.747	0.887	0.693	0.791	0.780
spellcheck	0.756 (+1.2%)	0.877 (−1.1%)	0.714 (+3.0%)	0.798 (+0.8%)	0.786 (+0.8%)
unicode	0.751 (+0.5%)	0.887 (+0.0%)	0.688 (−0.7%)	0.788 (−0.4%)	0.779 (−0.1%)
discretize	0.747 (+0.0%)	0.887 (+0.0%)	0.693 (+0.0%)	0.791 (+0.0%)	0.780 (+0.0%)
MV@3	0.730 (−2.3%)	0.776 (−12.5%)	0.577 (−16.8%)	0.754 (−4.7%)	0.709 (−9.0%)
MV@7	0.744 (−0.3%)	0.797 (−10.2%)	0.596 (−14.1%)	0.751 (−5.1%)	0.722 (−7.4%)
sc_mv@3	0.735 (−1.6%)	0.779 (−12.2%)	0.606 (−12.6%)	0.744 (−6.0%)	0.716 (−8.2%)
char_noise@0.1	0.728 (−2.5%)	0.755 (−14.9%)	0.645 (−7.0%)	0.730 (−7.7%)	0.715 (−8.3%)
MACABEU-off	0.755 (+1.1%)	0.886 (−0.2%)	0.713 (+2.8%)	0.806 (+1.8%)	0.790 (+1.3%)
MACABEU-oracle	0.741 (−0.8%)	0.870 (−1.9%)	0.690 (−0.5%)	0.798 (+0.8%)	0.775 (−0.6%)
MACABEU-estimated	0.743 (−0.5%)	0.829 (−6.6%)	0.628 (−9.5%)	0.794 (+0.4%)	0.748 (−4.0%)

Table 2: Clean accuracy per (defence, task), averaged over 3 victims. Each non-baseline cell: absolute accuracy and % change vs. none. Best non-baseline per column in **bold**.

Bug, spellcheck drops mean BODEGA from 0.259 (undefended) to 0.087 — a 66% reduction. Against the three word-level attackers (BERT-Attack, PWWS, Genetic) it is essentially inconsequential (mean BODEGA 0.448/0.366/0.344 vs. 0.445/0.357/0.376 undefended). The pattern is uniform across all victims.

(ii) **Output masking uniformly weakens all four attackers.** Hiding the victim’s soft probabilities (discretize) cuts mean BODEGA from 0.359 to 0.241 — a 33% reduction — and the per-attacker drop is similar in every column (BERT-Attack −34%, PWWS −27%, DeepWordBug −31%, Genetic −39%), with no input transformation applied.

(iii) **Ensembling dominates among single-action defences.** Majority-voting and its spell-corrected variant (MV@3, MV@7, sc_mv@3) all bring mean BODEGA below 0.14 across the four standard attackers, with sc_mv@3 the strongest at 0.123 mean. char_noise@0.10 achieves comparable robustness (0.138) at a higher clean-accuracy cost (Table 2).

Clean-accuracy cost. Table 2 reports clean accuracy on unattacked inputs, averaged across the three victims, with each non-baseline cell annotated by its percentage change relative to the undefended baseline. Input-perturbing defences pay the largest tax: char_noise@0.10 loses 8.3% relative to baseline (0.715 vs. 0.780), and the majority-voting variants lose 7–9%. spellcheck (+0.8%) and unicode (−0.1%) are accuracy-preserving, and discretize is identical to the baseline by construction (it touches only the output). The offline MACABEU policy, free to pick none on benign inputs, attains the highest mean clean accuracy

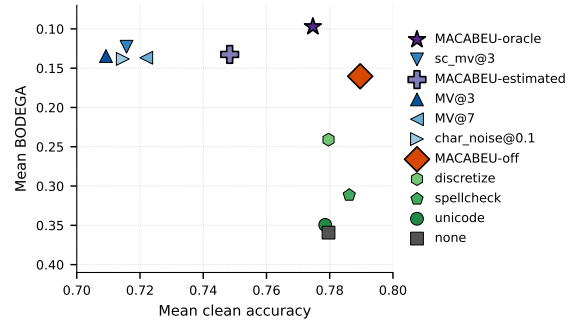


Figure 3: Accuracy–robustness trade-off. x : mean clean accuracy; y : mean BODEGA over the four standard attackers. Top-right is best.

(0.790, +1.3%). Figure 3 plots all eleven policies on the accuracy–robustness plane. MACABEU-off sits at the top-right corner: it yields the highest clean accuracy and cuts mean BODEGA by 55% relative to none (0.160 vs. 0.359). When no attack is present, the policy falls back to the accuracy-preserving defences in its action library; under attack, it switches to the more aggressive transforms that trade some accuracy for robustness. We explain this behaviour in Section 5.3. The two online variants push BODEGA further down at a modest accuracy cost: MACABEU-oracle to 0.097 (−1.5% accuracy vs. baseline), and MACABEU-estimated to 0.133 (−4.2% accuracy) — both Pareto-dominate the static ensemble defences (sc_mv@3 at 0.123 BODEGA and −8.2% accuracy).

Best static defences going forward. We select the strongest static defences from the aggregated grid for RQ2/RQ3. Among them, spellcheck_mv@3 is the single strongest defence on mean BODEGA and the natural target for an

Task	Pretrained		Retrained	
	none	mv@3	none	mv@3
PR2-BiLSTM	0.625	0.400	0.621	0.321
FC-BiLSTM	0.779	0.526	0.722	0.466
HN-BiLSTM	0.477	0.216	0.488	0.143
RD-BiLSTM	0.649	0.353	0.475	0.226
PR2-BERT	0.572	0.157	0.318	0.292
FC-BERT	0.773	0.512	0.645	0.386
HN-BERT	0.298	0.014	0.284	0.025
RD-BERT	0.497	0.261	0.346	0.192
Mean	0.584	0.305	0.487	0.256

Table 3: Mean BODEGA of pretrained vs. retrained adaptive attacker. Sub-cols: none (undefended) or mv@3 (spellcheck_mv@3). Lower = stronger; best per row in **bold**. Mean over fully evaluated rows only.

adaptive attacker.

5.2 RQ2: Adaptive attacks

Pretrained XARELLO vs. static defences. The rightmost column of Table 1 reports XARELLO’s mean BODEGA under three defences for which we have run the adaptive attacker on all 12 task–victim combinations. Against the undefended victim XARELLO scores 0.585, well above any standard attacker, confirming the gap between adaptive and non-adaptive search. Wrapping the same victim in spellcheck_mv@3 — the strongest static defence on the standard-attacker grid — only reduces XARELLO’s mean BODEGA to 0.374. By contrast, the standard attackers all collapse to ≤ 0.16 under the same defence: a $\sim 60\%$ gap between adaptive and non-adaptive damage at equal defence.

Retraining against a stochastic defence weakens the adaptive attacker. A defence in the adaptive threat model should be tested against an attacker retrained with the defence in place. Table 3 reports this: a second adaptive agent, retrained from scratch against spellcheck_mv@3, is evaluated alongside the pretrained one on both undefended and defended deployments. On the defended deployment the retrained agent is worse than the pretrained one in 6 of 8 task–victim combinations (mean BODEGA 0.256 vs. 0.305); only PR2-BERT and, marginally, HN-BERT improve — both combinations where the pretrained attacker was already crushed by the defence (< 0.2 BODEGA).

5.3 RQ3: Adaptive defences

MACABEU vs. adaptive attacker. Table 1 shows mean BODEGA in the XARELLO column: 0.585 undefended, 0.542 with MACABEU-off, 0.356 with MACABEU-oracle. Offline barely helps and is worse than spellcheck_mv@3 (0.374). Oracle cuts BODEGA by 39% vs. undefended and wins on all 12 task–victim combinations (Figure 1), 16–56% below the offline policy. **MACABEU-estimated reaches 0.378, only 6% above oracle — most of the online gain survives replacing the gold label with MV7.**

MACABEU vs. standard attackers. The first four columns of Table 1 tell the same story against the standard attackers. Online MACABEU has the lowest mean BODEGA in every column. The offline version uses the same eight actions and is beaten by sc_mv@3 in every column, so the win comes from updating during the attack, not from a stronger library.

MACABEU-oracle is a theoretical upper bound since it needs the gold label at inference time. Its deployment-realistic counterpart, MACABEU-estimated, uses a MajorityVote-7 pseudo-label instead and approaches the oracle on character-level attackers (BERTattack and XARELLO, within 5% of oracle); on word-level ones (PWWS, Genetic) it loses more ground (+60% and +27%). Estimated still beats every static defence. Appendix E has the per-attacker numbers.

What does the online policy learn? Figure 4 shows online MACABEU-oracle’s post-warmup action distribution by attacker (left) and by task (right). With no attacker, spellcheck accounts for 47% of selections and char_noise@0.1 for only 6%; under attack these flip. Within the attacker columns three patterns stand out. (i) char_noise@0.1 dominates against all four standard attackers (20–34%): each commits to a fixed search procedure, and a randomised input transform invalidates the importance scores it caches across queries. (ii) Against DeepWordBug — the only character-level attacker — spellcheck (20%) and unicode (18%) rise close to char_noise@0.1, since they specifically reverse typos and homoglyph-like edits. (iii) Under XARELLO the policy shifts away from randomised defences: spellcheck becomes the most-selected action (22%) and char_noise@0.1 drops to 16%. XARELLO’s Q -network averages many queries per example, so a stochastic transform feeds it noise but a deterministic cleaner cannot be averaged out.

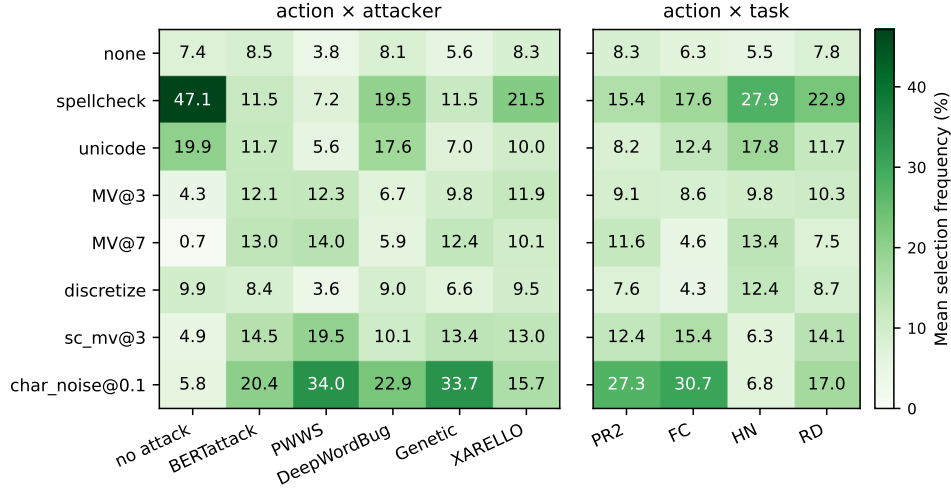


Figure 4: Online MACABEU’s mean action-selection frequency (%). *Left*: action × attacker. *Right*: action × task.

Tables 6–8 (appendix D) detail these results.

We also analyse which features MACABEU actually uses, which shows that MACABEU mostly relies on text length (in characters and words), as well as the ratio of uppercase and out-of-vocabulary words to choose the right defence. The details are in appendix F.

6 Discussion

Online MACABEU wins by adapting to the deployed attacker at inference time, not by using a stronger action library — it shares the same eight actions with the offline policy. The qualitative ranking of static defences is preserved across BiLSTM, BERT, and Gemma (ensembles dominate; character-level cleaning helps only against character-level attacks; output masking buys a constant fraction), and the online MACABEU win is victim-uniform (BERT −41%, Gemma −33%; the latter reflects Gemma being intrinsically harder to attack). MACABEU-oracle is a theoretical upper bound since it needs the gold label at inference time. MACABEU-estimated replaces that reward with a MV7 pseudo-label and approaches the oracle’s performance without any label access: mean BODEGA is only ∼20% higher on the BiLSTM+BERT sub-grid, and still below every static defence (Appendix E). The gap is larger on word-level attackers (PWWS +60%, Genetic +27%), where substitutions do not break the MV7 vote and the pseudo-label reinforces the wrong action.

Black-box preprocessing nonetheless has a hard floor: even under online MACABEU, XARELLO retains a residual attack success rate (ASR) of 15–76% across all combinations. When the at-

tacker produces semantically faithful, character-clean perturbations that genuinely flip the victim’s decision boundary, no input-side transformation can recover the original prediction. Closing the gap requires complementary defences — input authenticity, provenance, or victim-side modification — which fall outside our threat model. Online MACABEU’s per-query overhead (one Adam step on a 3k-parameter network) is negligible when compared against the victim forward pass.

7 Conclusion

We presented the first systematic comparison of black-box preprocessing defences for misinformation detection across four tasks, three victim architectures, and five attackers, and introduced MACABEU, an RL-based meta-defender that selects defences per example and keeps learning during the attack. Across the resulting 12 task–victim combinations, online MACABEU reduces an adaptive attacker’s BODEGA score by 16–56% relative to its offline counterpart, without modifying the victim. We release all code and policies in the hope that adaptive defence — cheap to retrofit onto already-deployed classifiers — becomes a routine companion to the systems it protects.

Limitations

(i) **Language coverage.** All experiments are restricted to English-language credibility datasets, and the SymSpell dictionary, BLEURT model, and pretrained victims are likewise English-specific. The MACABEU architecture itself is language-agnostic — the action library only requires per-

language analogues of the underlying transforms — so extending the study to multilingual misinformation corpora would be feasible.

(ii) Hand-curated action space. MACABEU’s eight actions are static, character- or token-level preprocessing primitives. Two natural directions for richer libraries are paraphrase- and purification-based defences, where an LLM rewrites the input (Moraffah et al., 2024; Robey et al., 2025), and learned token-substitution defences such as robust encodings (Jones et al., 2020) or adversarial typo correction (Pruthi et al., 2019). Because MACABEU’s selection is decoupled from the action implementations, such defences can be added as additional arms without retraining the meta-controller.

(iii) Single-attacker assumption. Online MACABEU is trained against a single attacker per session and assumes the attacker’s strategy is approximately stationary across queries. An adversary that switches attack styles mid-session, or that coordinates several attackers against the same defender, would induce a non-stationary environment for the online learner. Mitigations that maintain ensembles of frozen policies and switch between them at detected change points, or that lower the effective learning rate when reward variance spikes, are directly compatible with our architecture and a natural follow-up.

(iv) Compute scope. To keep total GPU usage tractable on a single-GPU SLURM cluster, we deliberately restricted two of the heaviest sub-grids. Gemma was evaluated only against the top eight static defences identified in Section 5.1, since each Gemma attack run is roughly an order of magnitude more expensive than its BERT or BiLSTM counterpart; the standard-attacker grid for the seven discarded defences is dominated by ensemble defences that are already weaker than `sc_mv@3`, so the omission does not affect the qualitative ranking. Retrained-XARELLO results are similarly restricted to BiLSTM (Table 3), where each retraining run is cheapest; the pretrained-XARELLO grid against MACABEU-oracle is complete across all 12 task–victim combinations. **MACABEU-estimated was evaluated on BiLSTM and BERT but not on Gemma. The MV7 pseudo-label reward needs seven extra base-victim forward passes per online update, which pushed each RD–Gemma cell past a single-GPU week. Appendix E therefore reports**

estimated on the BiLSTM+BERT sub-grid only.

(v) Semantic-similarity proxy. BODEGA uses BLEURT (Sellam et al., 2020) as its single semantic-similarity component. Alternative proxies (BERTScore, COMET, human ratings) could shift the absolute BODEGA scale; we expect the qualitative comparisons between defences to remain stable, but a robustness study across metric choices would strengthen the claim.

(vi) Black-box scope. Our threat model excludes any modification to the victim model, which is the right setting for retrofitting defences onto already-deployed classifiers but leaves complementary lines of work — adversarial training of the victim (Madry et al., 2018; Miyato et al., 2017), input authenticity checks, and provenance signals — outside the scope of this study.

(vii) Query budget. Each attacker is allowed up to 500 queries per example, which is higher than what a deployed misinformation classifier would typically expose: real platforms aggressively rate-limit queries and observe only a small fraction of them. Approaches that explicitly bound the attacker’s query budget, such as TREPAT (Przybyła et al., 2025), are a natural complement to adaptive preprocessing and would tighten the threat model used here.

(viii) Victim scale. Our victims — a BiLSTM trained from scratch, BERT-base fine-tuned in full, and Gemma-2B with QLoRA — are small by 2026 production standards. Real deployments increasingly rely on multi-billion-parameter LLMs, whose decision surfaces, vulnerability profile, and clean-accuracy budget under a given preprocessing defence may differ from what we observe at this scale. A follow-up evaluating MACABEU against larger victim backbones (Gemma-7B/27B, Llama-3 8B/70B) would be necessary to extrapolate our quantitative claims.

(ix) Fine-tuning vs. prompting. All three victims are fine-tuned task-specific classifiers. Prompting-based misinformation detection — where the same backbone is instructed in-context rather than fine-tuned on the BODEGA training split — is increasingly common in practice and exhibits a different robustness profile (in particular, prompt injection becomes a parallel attack surface). MACABEU is in principle backbone-agnostic, but its action library, learning rate, and warm-up bud-

get were tuned for fine-tuned victims and would have to be re-validated under prompting.

(x) Pseudo-label quality for online adaptation. MACABEU-estimated uses MajorityVote-7 on the defended input as the reward signal instead of the gold label. This is a good proxy for character-level attackers: the perturbations tend to break the attack and MV7’s vote lines up with the true label. It is a poor proxy for word-level attackers (PWS, Genetic): synonym substitutions survive the perturbation, so MV7 agrees with the fooled prediction and the reward pushes the policy the wrong way. Better pseudo-labels are a natural next step — for example, paraphrase or back-translation for an orthogonal signal, delayed user signals (report rates, moderation actions) in a deployed system, or human review triggered only on uncertain cases.

Ethical Considerations

Dual-use risk. The defences we study are explicitly designed to harden misinformation classifiers deployed against adversarial users. The same action library — spell-correction, Unicode normalisation, majority-vote averaging, character noise, output discretisation — is, however, equally in-spectable by an attacker who wishes to design perturbations that survive a given defence. We argue that the asymmetry favours defenders in this setting: the offline MACABEU policy and all static defences are already accessible to a black-box attacker through query observation (Section 5.2), so disclosure does not materially expand the attacker’s surface; conversely, the online MACABEU regime — which is the central contribution — is, by construction, only usable by an operator with control of the inference pipeline and therefore strengthens defenders disproportionately.

Misuse of misinformation detectors. Robust credibility classifiers can in principle also be used to over-filter legitimate dissenting speech. Our experiments measure robustness to adversarial perturbation, not the calibration of any particular deployment decision; the defences studied here neither shift the original classifier’s decision threshold (with the exception of discretize, whose effect is bounded and reported) nor change the set of inputs that the underlying victim labels as non-credible. Decisions about how to act on a classifier’s output remain the responsibility of the operator.

Reproducibility and AI assistance. We release all code, MACABEU policies, and configuration under a permissive licence, alongside the SLURM harnesses needed to reproduce the experiments on a single-GPU node. As disclosed in Appendix C, the implementation was developed with the help of LLM-based coding assistants; all generated code was reviewed, tested, and validated by the authors.

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1088	bustness to adversarial word substitutions . In <i>Pro-</i>	Task 11), FC (FEVER), HN (PAN @ SemEval-	1143
1089	<i>ceedings of the 58th Annual Meeting of the Associa-</i>	2019), and RD (augmented RNR). Sizes are total	1144
1090	<i>tion for Computational Linguistics, ACL 2020, On-</i>	examples per split, computed as exact line counts	1145
1091	<i>line, July 5-10, 2020</i> , pages 3465–3475. Association	from the released TSVs (no header rows). The	1146
1092	for Computational Linguistics.	% <i>pos.</i> column is the share of the positive (non-	1147
		credible) class on the <i>attack</i> split, which is the	1148
1093	Yuan Zang, Fanchao Qi, Chenghao Yang, Zhiyuan Liu,	subset actually evaluated: FC and HN are balanced	1149
1094	Meng Zhang, Qun Liu, and Maosong Sun. 2020.	by construction on that split, whereas PR2 and RD	1150
1095	Word-level textual adversarial attacking as combi-	remain skewed towards the credible class. <i>Mean /</i>	1151
1096	natorial optimization . In <i>Proceedings of the 58th</i>	<i>med. words</i> are whitespace-tokenised counts, also	1152
1097	<i>Annual Meeting of the Association for Computa-</i>		
1098	<i>tional Linguistics, ACL 2020, Online, July 5-10, 2020</i> ,		

on the attack split. The four tasks span more than an order of magnitude in both size and average input length — from PR2 sentences (~ 21 words) to HN articles (~ 605 words) — giving the defence evaluation a deliberately heterogeneous footprint.

Task	Train	Dev	Attack	% pos.	Mean / med. words
PR2	12,675	3,320	416	29.6	21 / 18
FC	172,763	19,010	405	50.6	39 / 31
HN	60,235	3,600	400	50.5	605 / 444
RD	8,694	2,070	415	32.8	253 / 164

Table 4: Per-task data statistics for the four BODEGA credibility tasks.

B Defence action gallery.

Table 5 shows the eight MACABEU actions applied to a single PR2 example, illustrating exactly what each defence does to the text seen by the victim classifier. Input transformations (spellcheck, unicode, majority_vote, spellcheck_mv, char_noise) are shown with the modified spans in **bold**; discretize acts only on the classifier’s output distribution, so the input is left unchanged.

C Implementation details

Hardware and tooling. All experiments are run on a single-GPU SLURM cluster (NVIDIA A40/L40s, 48 GB). Victims and attackers follow the BODEGA reference implementation (Przybyła et al., 2024b): BiLSTM victims are trained from scratch on the BODEGA training splits over 300-dimensional GloVe embeddings (Pennington et al., 2014); BERT-base (Devlin et al., 2019) is fine-tuned for 3 epochs; Gemma-2B (Team, 2024) is fine-tuned with QLoRA (Detters et al., 2023) (rank 16, $\alpha = 32$, 4-bit NF4 quantisation). All attackers are wrapped through OpenAttack (Zeng et al., 2021) with a 500-query budget per example. Semantic similarity uses BLEURT-20 (Sellam et al., 2020; Pu et al., 2021); character similarity uses normalised Levenshtein distance (Levenshtein, 1966) via the editdistance library; spellchecking uses SymSpell with the standard 80k-word English frequency dictionary; Unicode confusable mappings come from the homoglyphs library and the official Unicode confusables table. Code, configurations, and trained MACABEU policies are released alongside the paper.

Compute budget. Experiments are carried out on a shared SLURM cluster, with each job allocated a single NVIDIA A40 or L40s GPU (48 GB). The full evaluation comprises 4 tasks $\times$ 3 victim classifiers $\times$ 12 defences $\times$ 5 attackers, plus MACABEU offline training (one run per task–victim combination) and online policy evaluation against XARELLO. Victim fine-tuning (BERT, Gemma) and BLEURT-based BODEGA scoring dominate the per-job cost; per-action MACABEU updates contribute a negligible fraction. Across all completed runs reported here and in development, the project consumed approximately **3,400 GPU-hours** (1,776 GPU jobs).

Software versions. The experiments use Python 3.10.20 with PyTorch 2.7.1 (CUDA 11.8, cuDNN 9.1), Hugging Face transformers 4.46.3, tokenizers 0.20.3, and datasets 4.7.0. Adversarial attacks run through OpenAttack 2.1.1; semantic similarity uses bleurt-pytorch 0.0.1 (BLEURT-20 checkpoint) and bert-score 0.3.13; character similarity uses editdistance 0.8.1; and the static-defence library uses symspellpy 6.9.0 (English 80k-word frequency dictionary) and homoglyphs 2.0.4. Gemma fine-tuning relies on peft 0.18.1, bitsandbytes 0.49.2, and accelerate 1.13.0; numerical code uses NumPy 1.26.4.

Reporting and variability. All reported BODEGA scores correspond to single attacker runs with fixed random seeds across defence, attacker, task, and victim seeds; we do not average over multiple repetitions per (defence, attacker, task, victim) cell, given the size of the evaluation matrix. The MACABEU *online* numbers carry inherent run-to-run variance from the per-query replay buffer and ϵ -greedy exploration; the consistency of the 12/12 win pattern in Figure 1 suggests this variance is small relative to the offline–online gap, but a multi-seed study would be needed to quantify it.

MACABEU state features. The state vector encodes ten features of the current input:

1. Length, in characters.
2. Word count (whitespace-tokenised).
3. Average word length.
4. Out-of-vocabulary ratio: fraction of whitespace tokens not found in the SymSpell English

MACABEU action	Text seen by victim classifier (changes in bold)
none	The memo said the CHA possesses voluminous evidence, including, "Interview notes with House Members' Chiefs of Staff," and "Logon activity and computer access logs." Prosecutors have not brought [...]
spellcheck	The memo said the CHA possesses voluminous evidence, including, "Interview notes with House Members' Chiefs of Staff," and "Logon activity and computer access logs." Prosecutors have not brought [...]
unicode	The memo said the CHA possesses voluminous evidence, including, "Interview notes with House Members' Chiefs of Staff," and "Logon activity and computer access logs." Prosecutors have not brought [...]
discretize	<i>(input unchanged; only modifies the classifier output distribution)</i>
majority_vote@3	Teh memo sad te CHA possesses voluimnious evidcece , includingf , " Intervie nowtes with House Membrs' Chiefs of Staff," axnd " Logon activiyt and compupter access logs." Prosectorbs hav not brought [...] <i>(one stochastic sample shown; full action averages over k)</i>
majority_vote@7	Teh memo sad te CHA possesses voluimnious evidcece , includingf , " Intervie nowtes with House Membrs' Chiefs of Staff," axnd " Logon activiyt and compupter access logs." Prosectorbs hav not brought [...] <i>(one stochastic sample shown; full action averages over k)</i>
spellcheck_mv@3	Teh memo sad te CHA possesses voluimnious evidcece , includingf , " Intervie nowtes with House Membrs' Chiefs of Staff," axnd " Logon activiyt and compupter access logs." Prosectorbs hav not brought [...] <i>(one stochastic sample shown; full action averages over k)</i>
char_noise@0.1	<u>The</u> memo <u>said</u> <u>the</u> CHA possesses <u>voluminous</u> <u>evidence</u> , <u>including</u> , " <u>Interview</u> <u>notes</u> with House <u>Members'</u> <u>Chiefs</u> of Staff," <u>and</u> " <u>Logon</u> <u>activity</u> and <u>computer</u> access logs." Prosecutors <u>have</u> <u>not</u> brought [...]

Table 5: Defence action gallery: each row shows the text actually fed to the victim classifier when the corresponding MACABEU action is selected. Example drawn from PR2 (row 0). Tokens differing from the original are shown in **bold**; underlined characters were Unicode homoglyphs (e.g. produced by char_noise) that we have mapped back to their ASCII look-alike so the row renders under standard pdfL^AT_EX.

1240	frequency dictionary.	with the help of LLM-based coding assistants	1260
1241	5. Non-ASCII character ratio.	(Claude Code with Opus 4.6 and 4.7). All gen-	1261
1242	6. Uppercase character ratio.	erated code was reviewed, tested, and integrated by	1262
1243	7. Digit ratio.	the authors; experimental design, hyperparameter	1263
1244	8. Punctuation density.	choices, analyses, and the paper text remain our	1264
1245	9. Repeated-character ratio: fraction of length-3	own.	1265
1246	substrings consisting of three identical charac-		
1247	ters, normalised by input length.		
1248	10. Shannon entropy of the character distribution.		
1249	MACABEU hyperparameters. Offline: replay	D Qualitative examples	1266
1250	buffer 50k, batch 64, $\eta = 10^{-3}$ (Adam), target up-	Tables 6, 7, and 8 present one per-task qualita-	1267
1251	date every 200 steps, $\gamma = 0$, 200 training episodes	tive trace for each of the three victim architec-	1268
1252	per task–victim. Online: warm-up 50 examples at	tures (BiLSTM, BERT, Gemma). Each row con-	1269
1253	$\varepsilon = 0.5$, replay buffer 16, batch 16, one update	trasts the adversarial example XARELLO pro-	1270
1254	per query post-warmup. Initial ε in offline mode	duces against a weaker baseline (the undefended	1271
1255	anneals from 1.0 to 0.05 over training.	victim for BiLSTM; the offline MACABEU pol-	1272
1256	Use of AI coding assistants. The implemen-	icy for BERT/Gemma) with the example it pro-	1273
1257	tation — training and evaluation scripts, data-	duces against online MACABEU. Two patterns	1274
1258	conversion utilities, SLURM job harnesses, and	emerge consistently. (i) Against the weaker base-	1275
1259	plotting/table-generation tooling — was developed	line, XARELLO commits to a small number of	1276
		high-leverage word-level substitutions (e.g. “pres-	1277
		ident” → “held” in PR2, “cornered” → “kidnap”	1278
		in RD). (ii) Against online MACABEU, the same	1279
		edit budget fragments into many shorter, lower-	1280
		impact character-level perturbations spread across	1281
		the input. Together — more edits, smaller per-edit	1282
		impact — both semantic and character similarity	1283

drop, lowering BODEGA on every task–victim combination.

E MACABEU variants comparison

Section 3.3 defined two online MACABEU variants: MACABEU-oracle, which uses the gold label to compute the reward, and MACABEU-estimated, which uses a pseudo-label from `majority_vote@7` on the defended input. Table 9 compares the two on BiLSTM and BERT, per attacker. We leave Gemma out because the estimated variant did not finish on the Gemma sub-grid in time — each of the three RD–Gemma cells needs about 140 hours of wall-clock, driven by the seven extra forward passes per online update.

Two things stand out. First, MACABEU-estimated keeps most of the offline-to-online gain: mean BODEGA drops from 0.244 (off) to 0.182 (estimated), and estimated still beats every static defence on the same sub-grid. Second, the gap to MACABEU-oracle is +20% on average, but it is not uniform: character-level attackers (BERTattack, XARELLO) stay within 5% of oracle, while word-level attackers (PWWS, Genetic) are +60% and +27% respectively. The reason is how MV7 behaves: its random character-level perturbations tend to break character-level attacks, so its vote is close to the true label. Word-level substitutions survive the perturbation, so the vote often matches the fooled prediction and the reward pushes the policy toward the wrong action.

F Feature importance analysis

Section 3.3 lists the 10 features MACABEU uses to pick a defence. To see which ones actually matter, we swap each feature’s value in a given example for the dataset mean of that feature, and check whether the greedy action changes. A feature is important if this swap flips the action often. We report the *flip rate* per feature: the fraction of examples where the action changed after the swap.

We run this on every trained policy on disk: one offline policy per (task, victim), and one online oracle and one online estimated policy per (task, victim, attacker). This gives 880 (policy, feature) flip rates in total.

Figure 5 averages the flip rate across all policies within each variant. Four features do most of the work: *text length* (0.18–0.26 flip rate), *uppercase ratio* (0.13–0.26), *word count* (0.13–0.22), and *OOV ratio* (0.14–0.23). The other six stay below

0.11 and barely influence the choice.

Two smaller findings. The offline policy has roughly double the flip rates of either online variant on the four dominant features — the frozen policy leans hard on single features, and online adaptation spreads the decision across more of them. And MACABEU-oracle and MACABEU-estimated agree on the per-feature flip rates within 0.02 on average, so the MV7 pseudo-label ends up using the same features as the gold label, just with a weaker reward signal.

Figure 6 breaks this down by attacker for the two online variants. The four dominant features stay dominant everywhere. DeepWordBug pushes text length higher on the long-text tasks (RD, HN), because character-level insertions inflate the input. XARELLO spreads its edits more, so the policy against XARELLO uses the four features more evenly.

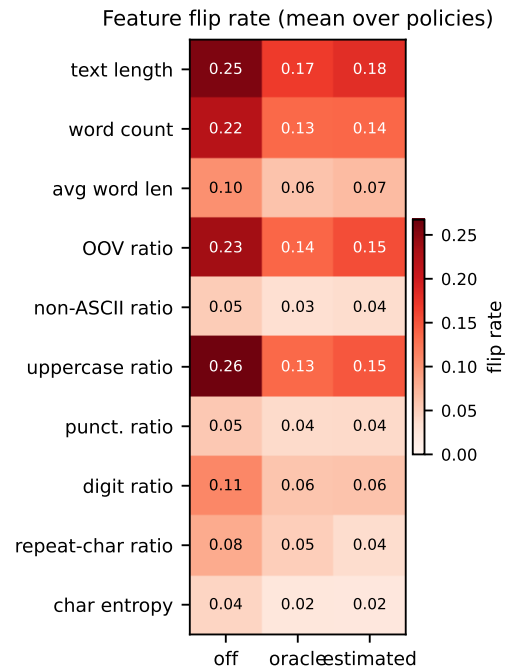


Figure 5: Mean flip rate per feature (rows), aggregated over all trained policies within each MACABEU variant (columns). Darker cells indicate features the policy relies on more heavily.

Task	Original	XARELLO (no defence)	XARELLO vs. online MACABEU
PR2	abid awan , a member of a family of highly paid democratic it aides , was awarded a \$ 45 , 000 payout in court after he enlisted a high profile attorney specializing in money [...]	abid adwan , a member of a family of highly paid democratic it aides , was awarded a \$ 45 , 000 payout in court after he enlisted a high profile attorney specializing in money [...]	aid adwan , a member of s family of highly paid democratic it aides , was awarded a \$ 45 , 000 payout in court after he enlisted a high profile attorney specializing in money [...]
FC	[...] books . ~ clueless (film) was preceded by a series of books .	[...] books . ~ clueless (film) : series , a series and books .	[...] books . ~ clueless (film) : series by a series of books .
HN	undercover : university of houston faculty members approve ridiculous emotional first aid kit concept in this hilarious and satirical video , james o ' keefe once again takes on [...]	un ver cover : university of houston faculty members approve ridiculous ex otional first aid kit concept in this hilarious and satirical video , james o ' keefe once again takes on [...]	un ver cover : university of houston faculty members approve ridiculous ex otional first aid kit concept in this hilarious and satirical video , james o ' keefe once again takes on [...]
RD	charlie hebdo suspects cornered with a hostage , want to die as " martyrs " http :// t . co / udgqozjank http :// t . co / prhci9nn5h keep it lite ... @ laurenonizzle @ [...]	charlie hebdo get cornered with a kidnap , want to die as " martyrs " http :// t . co / udgqozjank http :// t . co / prhci9nn5h keep it lite ... @ laurenonizzle @ [...]	charlie hebdo suspects cornered with a hostage , want to death as " saintss " http :// t . co / udgqozjank http :// t . co / prhci9nn5h keep it lite ... @ laurenonizzle @ [...]

Table 6: Qualitative comparison on BiLSTM: original text, the adversarial example XARELLO produces against the undefended classifier, and against an online MACABEU defender. Within-word character edits are shown in **bold**; whole-word substitutions in **bold + underline**.

Task	Original	XARELLO vs. offline MACABEU	XARELLO vs. online MACABEU
PR2	this would actually be the perfect time for trump not to appoint an ambassador to the un and call for the us to exit the anti - american organization and demand congress stop [...]	this would actually be the convenient time for trump not to appoint an ambassador to the un and call for the us to exit the anti - american organization and demand congress stop [...]	this would actually be the best time for trump not to appoint an ambassador to the un and call for the us to exit the anti - american organization and demand congress stop funding [...]
FC	[...] 1968 . ~ ernest medina was uninvolved in the my lai massacre .	[...] 1968 . ~ ernest medina was as involved in the my lai massacre .	[...] 1968 . ~ ernest medina was as alisted in the my lai massacre .
HN	vote : who is the real republican ' frontrunner ' for president ? revive america & # 160 ; is conducting an urgent poll of conservative voters about the 2016 republican [...]	apote : who is the real republican ' frontrunner ' for president } revive america & # 160 ; is conducting an urgent poll of conservative voters about the 2016 republican [...]	apoce : who is the rocl republican ' frontrunner ' for president } revive america & # 160 ; is conducting an urgent poll of conservative voters about the 2016 republican [...]
RD	3 suspects (34 , 32 , & 18 yrs old) have been identified in the # charliehebdo attack & are from reims (paris suburb) . 2 of them are brothers @ katunews say it with me ... [...]	3 victims (34 , 32 , & 18 yrs old) have been identified in the # charliehebdo murder & are from reims (paris suburb) . 2 of them are brothers @ katunews say it with me ... [...]	3 victims (34 , 32 , & 18 yrs old) have been identified in murder murder charlieheb murder murder & are from reims (paris suburb) . 2 of them are brothers @ katunews say it [...]

Table 7: Qualitative comparison on BERT: original text, the adversarial example XARELLO produces against the offline MACABEU defender, and against an online MACABEU defender. Within-word character edits are shown in **bold**; whole-word substitutions in **bold + underline**.

Task	Original	XARELLO vs. offline MACABEU	XARELLO vs. online MACABEU
PR2	[...] . s . president ." minhas is in prison for an unrelated fraud charge .	[...] . s . president . " minhas is in held for an unrelated fraud charge .	[...] . president . " minhas is in held for an unrelated us charge .
FC	[...] and occasional writer . ~ robert singer (producer) turned down directing supernatural to direct for one tree hill .	[...] and occasional writer . ~ " singer (producer) turned but directing supernatural to direct for . by n .	[...] and occasional writer . ~ " singer (producer) turned but directing supernatural to direct for one tree n .
HN	fake news media reports non - documented russian allegations against trump – we have actual documentation on the crimes & corruption of hillary clinton if you are wondering why we [...]	fake news media reports non - documented russian all liess against trump – we have actual more umentation on the crimes & corruption of hillary clinton if you are wondering why we [...]	fake news media reports non - documented russian all liess against trump – we have actual fileume file on the liess & corruption of hillary clinton if you are wondering why we [...]
RD	at least two killed in hostage drama east of paris http : // t . co / oturcuuhqq @ capitalfm _ kenya may god rescue them @ capitalfm _ kenya france , what is happening in france	atly two killed in terror drama east of paris http : // t . co / oturcuuhqq @ capitalfm _ kenya may god rescue them @ capitalfm _ kenya france , what is happening in france	at least are killed at hostage rape south paris http : // t . co / oturcuuhqq @ capitalfm _ kenya may god rescue them @ capitalfm _ kenya france , what is happening in [...]

Table 8: Qualitative comparison on GEMMA: original text, the adversarial example XARELLO produces against the offline MACABEU defender, and against an online MACABEU defender. Within-word character edits are shown in **bold**; whole-word substitutions in **bold + underline**.

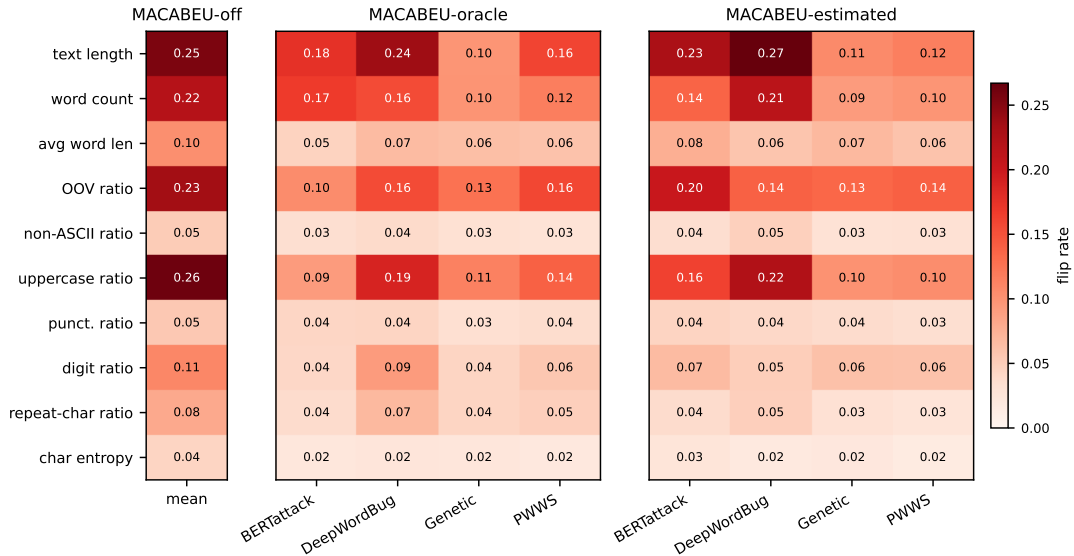


Figure 6: Per-attacker breakdown of feature flip rates. *Left*: MACABEU-off (a single column, since the offline policy is attacker-independent). *Middle*: MACABEU-oracle, one column per attacker. *Right*: MACABEU-estimated. Flip rates are averaged over the 4 tasks and 3 victim classifiers.

Attacker	off	oracle	estimated	est. vs. oracle
BERTattack	0.188	0.106	0.111	+4.9%
PWWS	0.174	0.110	0.175	+60.1%
DeepWordBug	0.107	0.068	0.105	+54.3%
Genetic	0.187	0.110	0.139	+27.2%
XARELLO	0.566	0.363	0.378	+4.3%
Mean	0.244	<u>0.151</u>	0.182	+20.3%

Table 9: Comparison of MACABEU variants (mean BODEGA on BiLSTM + BERT, averaged over 4 tasks). Lower = stronger defence; best per row in **bold**. The last column reports the relative change of the deployment-realistic **MACABEU-estimated** variant against **MACABEU-oracle** (which requires the gold label). GEMMA is excluded because the estimated variant did not finish on the GEMMA sub-grid within the revision cycle; the full grid is reported in Table 1 for MACABEU-off and MACABEU-oracle.